PANJABUNIVERSITY,CHANDIGARH-160014 (INDIA)

(Estd.underthePanjabUniversityActVIIof1947 –enacted by the Govt. of India)

SYLLABI

CourseOutlinefor
Bvoc (Nutrition & Dietetics)

as per NEP 2020

w.e.f session2024-25

Course Outline for Bvoc(Nutrition & Dietetics)
as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 101	Core Discipline	Human Physiology	2	2	4	40	10	40	10	100
NAD 102	Core Discipline	Basics of food and nutrition	2	2	4	40	10	40	10	100
NAD 103	Core Discipline	Indigenous Food systems	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	Common Basket			3					75
IDC I	Interdisciplinary Course	Common Basket			3					75
VAC I	Value-Added Course	Common Basket			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 201	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
NAD 202	Core Discipline	Food Preservation	2	2	4	40	10	40	10	100
NAD_203	Core Discipline	Nutrition and Immunity	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	Common Basket			3					75
IDC II	Interdisciplinary Course	Common Basket			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Food Microbiologist

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter2ndSemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in Nutrition & Dietetics = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a

UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 301	Core Discipline	Nutritional Biochemistry	2	2	4	40	10	40	10	100
_NAD 302	Core Discipline	Nutraceutical and Food additives	2	2	4	40	10	40	10	100
_NAD 303	Core Discipline	Food science	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 401	Core Discipline	Human Nutrition	4	-	4	80	20	-	-	100
_NAD 402	Core Discipline	Institutional Food Service & Personnel Management	2	2	4	40	10	40	10	100
_NAD 403	Core Discipline	Meal management	2	2	4	40	10	40	10	100
NAD 404	Core Discipline	Food Quality Management	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Nutrition educator

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter4th SemesterExaminations during summer break. Training

report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rdsemester.

** Minimum Credits Required to get Diploma Course in Nutrition & Dietetics = 96 + 4 #.

#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 501	Core Discipline	Therapeutic Nutrition-I	2	2	4	40	10	40	10	100
NAD 502	Core Discipline	Food Hygiene & Sanitation	4	-	4	80	20	-	-	100
NAD 503	Core Discipline	Food standards & Quality Control	4	-	4	80	20	-	-	100
NAD 504^	Core Discipline	Community Nutrition-I	2	2	4*	40	10	40	10	100
NAD 505^	Core Discipline	Health, Fitness and Sports Nutrition	2	2	4*	40	10	40	10	100
NAD 506^	Core Discipline	Diet Counselling	-	4	4*	-	-	80	20	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 601	Core Discipline	Therapeutic Nutrition II	2	2	4	40	10	40	10	100
_____NAD 602	Core Discipline	Introduction to Research Methodology and Report Writing	4	-	4	80	20	-	-	100
NAD 603	Core Discipline	Total Quality Management	4	-	4	80	20	-	-	100
NAD 604	Core Discipline	seminar	-	4	4	-	-	80	20	100
NAD 605^	Core Discipline	Community Nutrition-II	2	2	4*	40	10	40	10	100*
NAD 606^	Core Discipline	Clinical Assessment and Patient Counselling	2	2	4*	40	10	40	10	100*

NAD 607^	Core Discipline	Maternal & Child Health and Nutrition	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

Job Role – Clinical diet counselor

Semester-VII

Paper Code	**Type of Course**	**Subject Name**	**Credits**			**Marks**				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 701	Core Discipline	Advance nutrition	4	-	4	80	20	-	-	100
NAD 702	Core Discipline	Dietetics and Therapeutic Nutrition	2	2	4	40	10	40	10	100
_NAD 703	Core Discipline	Dissertation - I	-	4	4	-	-	80	20	100
NAD 704^	Core Discipline	Public Health and Nutrition	2	2	4*	40	10	40	10	100*
NAD 705^	Core Discipline	Principles of food processing	2	2	4*	40	10	40	10	100*
NAD 706^	Core Discipline	Institutional Food Administration	4	-	4*	80	20	-	-	100*
SIT 701	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

^Any two in 7th Semester

Semester-VIII

Paper Code	**Type of Course**	**Subject Name**	**Credits**			**Marks**				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 801	Core Discipline	Advanced Nutritional Biochemistry	2	2	4	40	10	40	10	100
_NAD 802	Core Discipline	Clinical Nutrition	2	2	4	40	10	40	10	100
NAD 803	Core Discipline	Women Health and Hygiene	4	-	4	80	20	-	-	100
NAD 804	Core Discipline	Dissertation- II	-	4	4	-	-	80	20	100
NAD 805^	Core Discipline	Extension Education in Nutrition	4	-	4*	80	20	-	-	100*
_NAD 806^	Core Discipline	Techniques in food analysis	2	2	4*	40	10	40	10	100*
NAD 807^	Core Discipline	Food service management	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 8th Semester

Job Role –Human Resource Officer

* SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter6th SemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 7th Semester. Viva-Voce examination to be held within 3 weeks of the start of 7thsemester.

PROGRAMMEOUTCOMES

PO1. Prepare qualified entry level professionals by providing expected domain knowledge.

PO2. Prepare a socially committed individual having high ethical values and empathy for the needs of society.

PO3. Design/development of solutions Develop innovative food products or substitutes or alternate solutions to create value and wealth for the betterment of the individual and society at large.

PO4. Problem analysis Identify, formulate, research literature, and solve nutritional deficiencies using fundamentals of clinical nutrition and dietetics, physiology, food science and biochemistry and relevant domain disciplines.

PO5. Professional Ethics Hold up and commit to professional ethics and ethical regulations, responsibilities, and norms of professional nutrition and dietetics practice.

PO6. Graduates are trained to develop innovative ideas and teams work skills to accomplish a common goal.

PO7. Produce lifelong learner with scientific skills for industrial applications and entrepreneurship.

Semester I

1	Course Title	Human Physiology
2	**CourseCode**	**NAD 101**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Exercise Physiologist
8	**Course Objectives**	1. To provide a basic introduction to human physiology: the study of the functions of the human body and its constituent parts. **2.** Apply knowledge of functional mechanisms and their regulation to explain the underlying physiology of common disease.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. This course is based on basics of Anatomy and Physiology highlighting structural anatomy of various human body organs. CO2. To understand how these separate systems interact to yield integrated physiological responses. CO3. Students will be able to associate the structural anatomy with functions of body organs. CO4. The focus of this course is to evaluate the effect of alterations in structure on the functions of organs. CO5. The students under this course will able to understand the working of equipment's and instruments used in physiology lab. CO6. The course gives an insight about various parameters for diagnosis of illness.
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

• Cell structure: Function and multiplication. Introduction to cell physiology and cell membrane
• Tissues: Types, structure, and their functions.
Circulatory system: Anatomy and functions. Heart rate, cardiac cycle. Cardiac output. Blood composition and functions. Blood clotting, blood groups, radial pulse, blood pressure and factors affecting it.
• Lymphatic system: Lymph glands and its function, spleen-structure, and functions.

UNIT– II

• Skeletal System: Bones, joints (in brief).
• Digestive system: Anatomy and functions of alimentary tract and accessory organs. Digestion and absorption of food.
- Respiratory system: Anatomy, functions, mechanisms of external and internal respiration

UNIT– III

• Endocrine System: List of endocrine glands, their hormones, functions, and their disorders (in brief)- Thyroid, parathyroid, adrenal, pancreas, pituitary.
• Urinary system: Anatomy and functions. Formation and composition of urine.

UNIT-IV

• Skin: Structure and functions
• Reproductive system: Structure and functions of male and female reproductive organs. Menstruation and puberty,
• Sensory organs: Eye, Ear, Nose, Tongue, skin – their function and basic structure (in brief).

Practicalwork:

1. Study of different types of microscopes and their uses.
2. Microscopic examination of permanent slides.
3. Identification of different blood groups.
4. Measurement of pulse and blood pressure.
5. Estimation of haemoglobin, RBC, WBC, TLC, DLC and ESR.

13 Suggested Resources for Reading

1.Keele, C.A. and Neil, E. Samson Wright ‘s Applied Physiology, Oxford University Press.
2. Tortora, G.J. and Anagnostakos, N.P. (1984). Principles of Anatomy and Physiology, Harper and Row Publisher.
3. Best, C.H. and Taylor, N.B., The Living Body, Asia Publishing House
4. Guyton, A.C., Functions of the Human Body, W.B. Saunders Co.
5. Wilson, K.J.W., Anatomy and Physiology in Health and Illness, ELBS/Churchill Livingstone.
6. VidhyaRatan, Handbook of Human Physiology, New Central Book Agency

Note: Latest edition of text books may be used.

1	Course Title	Basics of Food and Nutrition
2	**CourseCode**	**NAD 102**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritionist
8	**Course Objectives**	1. To elucidate the applied concepts of different areas of Foods and Nutrition. **2.** To acquire knowledge about the nutritional needs and concerns of specific age groups/physiological conditions.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. The course focuses on the functions of food for healthy life. CO2. To designed to make the students understand different aspects of nutrients. CO3.To designed to make the students understand different aspects of nutrients. CO4. The students will be able to understand the nutrition information based on scientific reasoning for clinical, community, and food service applications. CO5. This practical course is designed make the students understand the importance of weights and measures in nutritional cookery. CO6. It is designed to know the nutritional value of basic recipes of each category through calculations.
10	**Pedagogy of the Course Work**	**ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.**
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Introduction to nutrition: Food as a source of nutrients, food pyramid, function of foods, definition of nutrition, nutrients, and energy; adequate, optimum and good nutrition, malnutrition.
- Food guide – Basic five food groups and the nutritional contribution of each group in the diet.
- Interrelationship between nutrition & health -Visible symptoms of good health

UNIT– II

- Carbohydrates – Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body. Role of fibre in human nutrition.
- Fats & Oils - Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.

UNIT– III

- Proteins- Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.
- Water–role as a nutrient, functions, sources, requirement, water balance & effect of deficiency.
- Vitamins – their functions, sources, requirements, effect of deficiency and excess of following vitamins-
  a) Fat Soluble - Vitamins A, D, E and K
  b) Water Soluble- Ascorbic acid and B-Complex: Thiamine, Riboflavin, Niacin, Pantothenic acid, Folic acid and B12.

UNIT-IV

- Minerals –their functions and sources. Bioavailability, requirements, and effect of excess and deficiency of calcium, iron, iodine, sodium, potassium, and fluorine.
- Functional foods.
- Importance and methods of cooking and their effect on the appearance, nutritional value & taste**.**

Practicalwork:

1. Use and Care of different Kitchen equipment's
2. Weights and measures-Standard, Household measures for raw and cooked food.
3. Preparations from cereals (ANY TWO)- e.g. boiled rice, rice, pulao, chapati, puri, paratha, sandwiches, pastas, pancakes, cookies, and cake.
4. Preparations from pulses (ANY TWO)- e.g. dal makhani, sambar, chana masala, rajma etc.
5. Preparations from vegetables (dry, curries, koftas) (ANY TWO)- e.g. mixed vegetable, gobhi masala, vegetable kofta etc.
6. Preparations from milk (ANY TWO)- e.g. Porridge, curd/raita, paneer dishes, desserts and puddings custard, kheer, ice cream.
7. Soups (ANY TWO)- Basic, clear, and cream soups.
8. Salads with and without dressing. (ANY TWO)
9. Beverages (ANY TWO)- preparation of nourishing and refreshing beverages.
10. Snacks (ANY TWO) – pakoras, cutlets, cheese toast, upma, poha, peanut chikki, til-laddu preparations.
11. Use of leftover foods.

Requirements of all the nutrients should be according to RDA by NIN, ICMR

13 Suggested Resources for Reading

1. Guthrie, H., Andrews, Introductory Nutrition, St. Louis, Mirror /Mosby Collage.
2. Mudami S.R. &M. VRajagopal, Fundamentals of Food and Nutrition, Wiley Eastern
3. Swaminathathan, M., Advanced Textbook of Foods, and Nutrition, Vol I, II, Bappco, Banglore.
4. Wilson, E.D., Principals of Nutrition, John Wiley &Sons
5. Swaminathathan, M, Principals of Nutrition and Dietetics, Bangalore Print & Publishing Co.
6. William, S., Nutrition and Diet Therapy, Mosby Co, St. Louis.
7. Bogert, L.J., & Calloway, D.H., Nutrition and Physical Fitness, W.B. Saunders & Co, Philadelphia.

Note: Latest edition of text books may be used.

1	Course Title	Indigenous Food System
2	**CourseCode**	**NAD 103**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Ayurvedic physicians
8	**Course Objectives**	1. To impart knowledge regarding promotion and conservation of medicinal plants **2.** Apply knowledge regarding Ayurvedic medicinal plants and its impacts on health.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. This course describes both broad knowledge and in-depth relation of herbs and nutrition. CO2. It is based on study of different herbs and their use in day-to-day life. CO3. The course also relates nutrition and functional components of herbs by exploring the newer product development.
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Ayurveda: Concept, history, classification and uses
- Basic principles of Ayurveda, Allied systems of medicine in
- Ayurveda. Concept of diet in Ayurveda.
- Status of Ayurveda and Perspectives for its future application.

UNIT– II

- Introduction to medicinal plants- Aloe Vera, Tulsi, Neem, Turmeric, Ginger, Moringa, Amalta and Brahmi; functional properties, traditional practices and nutritional significance Promoting sustainability, conservation, and utilization of medicinal plants.

UNIT– III

- Role of herbs and spices in developed and developing countries.
- Utility of medicinal plants in household remedies of curing various diseases
- Functional Foods grown at household level and its role on human health; Factors influencing use of food for medicinal purposes.

UNIT-IV

- Medicinal Plant market and industries in India.
- Medicinal food products of the future - novel foods

13 Suggested Resources for Reading

1. Joshi, M.C., Text book of Indian Medicinal Plants –Scientific Publishers
2. Bhutya, R K., Handbook of Ayurvedic Medicinal Plants of India-Scientific Publishers
3. Kapoor, L.D., Handbook of Ayurvedic Medicinal Plants

Note: Latest edition of text books may be used.

SEMESTER – I

PW I
Project Work

- Survey to evaluate the dietary habits and assess the nutritional status of adolescent girls and middle-aged women, followed by an analysis and recommendation.
- Proposing healthy and nutritious recipes tailored for these demographic group.
- Submitting of project report.

Semester II

1	Course Title	Food Microbiology
2	**CourseCode**	**NAD 201**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+10IA) + (40 PRACTICAL + 10 IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Microbiologist
8	**Course Objectives**	1. To study the origin of micro-organisms and their structure. 2. To understand the growth and activities of bacteria. 3. To gain knowledge of principles of various techniques used in prevention and control of micro- organisms in food. **4.** To develop the knowledge about harmful bacteria, methods of transmission and anti-microbiology agents: antibiotics, germicides
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Explain the interactions between microorganisms and the food environment, and factors influencing their growth and survival. CO2. Explain the significance and activities of microorganisms in food. CO3. Describe the characteristics of foodborne, waterborne and spoilage microorganisms, and methods for their isolation, detection, and identification. CO4. Explain why microbiological quality control programmes are necessary in food production.
10	**Pedagogy of the Course Work**	ClassRoomLecture,Laboratory,Interactivesessions,PPT,PracticalandGroup Discussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

Introduction of microbiology and its relevance to everyday life, general characteristics of bacteria, fungi, virus, protozoa and algae.

Growth of microorganisms: Growth curve, effect of environmental factors in growth of microorganism, pH, water activity, oxygen availability, temperature, and others.

UNIT– II

Microbiology of different foods: Spoilage and contamination, sources, types, effect on the following:

(a) Cereal and cereal products
(b) Sugar and sugar products
(c) Vegetables and fruits
(d) Meat and meat products
(e) Fish, egg, and poultry
(f) Milk and milk products
(g) Canned foods.

UNIT– III

Environmental microbiology:

(a) Water and water borne diseases
(b) Air and air borne diseases
(c) Soil and soil borne disease
(d) Sewage and diseases

Beneficial effect of microorganisms.

Fermented foods: Bread, malt beverages.

UNIT– IV

Role of microorganisms in fermented milk products: Butter and cheese etc.

Microbial enzymes in food processing.

Relevance of microbial standards for food safety. Food hazard of microbial origin and occurrence and growth of microorganism in food.

Microbial intoxication and infections: Sources of contamination of food by pathogenic organisms, symptoms, and methods of control.

Practicalwork:

1. Study of equipment's in a microbiology lab.
2. Preparation of laboratory media and special media, cultivation and identification of bacteria, yeasts, and moulds.
3. Gram staining of bacteria.
4. Microbial examination of various foods: Foods and vegetables, meat, milk, and milk products.
5. Dye reduction test for assessing quality of milk.
6. Sterilization of various utensils.
7. Testing of water quality.
8. Development of aseptic conditions.
9. Visits to food processing units or any other organization dealing with advanced methods in food microbiology.

13 Suggested Resources for Reading

1. Frailer, W.C., Food Microbiology, McGraw Hill, New York
2. Kawata,K. Environmental Sanitation in India, Lucknow Publ. House
3. Pelezer, H.J. and Rober,D., Microbiology, McGraw Hill, New York
4. Banwart, G.T., Basic Food Microbiology, CBS Publ., New Delhi
5. Jay,J.H., Modern Microbiology, CBS Publ., New Delhi

Note: Latest edition of text books may be used.

1	Course Title	Food Preservation and Storage
2	**CourseCode**	**NAD 202**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY+10IA) + (40 PRACTICAL + 10 IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food technologists
8	**Course Objectives**	1. To impart knowledge on the causes of food spoilage and methods of processing and preserving food. 2. To identify & select processing equipment and preservation methods appropriate for specific foods. **3.** To describe the effects of methods of preservation on the quality of food.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. To understand the basic principles of food preservation CO2. To discuss the qualities and standards of available food preservatives and their suitability for different purposes. CO3. In this portion students learn about various techniques of preserving food products and detailed study of preserved food products, storage, nutritional aspects and food standards and food laws. CO4. students learn different methods of Food preservation – Drying, Freezing, Frying, canning, bottling etc. and preparation of jam, jelly, pickles, sauces, squashes.
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Definition, scope, and importance of food preservation.
- Classification of food based on its shelf life.
- Food spoilage: cause and effects.
- Principles of Food preservation
- Techniques of food preservation

UNIT– II

- Methods of food preservation-Canning: Principles of Preservation of canned foods.
- Thermal process for canned foods, spoilage in canned foods.
- Water activity of food and its significance in food preservation;
- Dehydration and drying of food items;
- Low temperature preservation: cold storage, cold chain, freezing (including cryogenic freezing)
- Preservation by fermentation: curing and pickling;
- Non-thermal (e.g. high-pressure processing)
- Ionization radiation; Use of preservative in foods:
- Chemical preservative, bio preservatives, antibiotics, lactic acid bacteria

UNIT– III

- Selection and purchase of foods for preservation.
- Storage of common perishable, semi-perishable and non-perishable foods.
- Storage stability of selected Frozen Foods.

UNIT-IV

- Packaging and packing materials, labelling, and costing of products.
- Food additives, preservatives, anti-oxidants, colouring matter, and emulsifying agents.

Practicalwork:

1. Preparation and storage of:
    a) Squashes and syrups
    b) Jams and jellies
    c) Marmalades and other fruit preserves
    d) Ketchup/sauces and chutneys
    e) Pickles (with and without oil).
2. Preparation of items utilizing cereals and legumes and their storage.
3. Demonstration of canning and bottling of fruits and vegetables.
4. Demonstration of Sealing Process used in Food Industry
5. Visit to food industry.

13 Suggested Resources for Reading

1. Desrosier, L.P, The Technology of Food Preservation, CBS Publishers & Distributors
2. Potter, N.N & Hotchkiss, J.H., Food Science, CBS Publishers & Distributors
3. Srivastava, R.P. & Kumar, S., Fruits and Vegetable Preservation, Principle and Practices, CBS Publishers & Distributors
4. Muller, H.G., Fundamentals of Foods and Nutrition. Wiley Eastern Pvt Ltd.
5. Modern Technology on Food preservation by NIIR Board.

Note: Latest edition of text books may be used.

1	Course Title	Nutrition & Immunity
2	**CourseCode**	**NAD 203**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutrition consultant
8	**Course Objectives**	Examining how nutrition reduces the side effects of medical treatments in general.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. To provide knowledge about various foods and their quality and how to use them to provide immunity to the human body. CO2. The immune system can boost some components, such as vitamins, minerals, antioxidants, and phenolic compounds, because these components aid in forming T cells, B cells, and antibodies and enhance the function of melanoproteins and neutrophils. CO3. To understand the role of functional food ingredients of various foods and their mechanism. CO4. To illustrate the health benefits of using nutraceuticals
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Definition of immunity, Immune system: The defence mechanisms of the body.
  - Role of nutrients in immune functions. Under nutrition-causes and consequences.

UNIT– II

•Infection: Risk of infection, influence of infections on nutrition and susceptibility to infection.

• Role of nutrients in prevention of risk of infections.

• Childhood mortality associated with infectious diseases of gastro-intestinal tract and lungs.

UNIT– III

• Age related immune depression.

• Role of nutraceuticals and functional foods.

• Nutrition in HIV/AIDS and tuberculosis.

UNIT-IV

• Probiotics, prebiotics, and immunity

• Food allergy (with emphasis on wheat allergy, peanut, lactose intolerance)

- Latest health concerns

13 Suggested Resources for Reading

1. Barasi, M. E., Human Nutrition-A health perspective. Arnold International Student ‘s edition.
2. Gershwin, M.E., Nestel, P. and Keen, C.L., Handbook of Nutrition and Immunity. Humana Publishers
3. Owen, J., Punt, J. and Stanford, S., Kuby Immunology: International Edition. Macmillan Publishers.
4. Prason, S.K. and Poddar, T., Perfect Health-Body Diet and Nutrition. V & S publishers.

Note: Latest edition of text books may be used.

SEMESTER III

1		**Nutritional Biochemistry**
2	**Course Code**	**NAD 301**
3	**Typology ofCourse**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritional Biochemist
8	**Course Objectives**	To lay the foundation for an introduction to various bio-molecules and understanding of various metabolic pathways**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1**: Compare different metabolic pathways of carbohydrates. **CO2**: Distinguish between different metabolic pathways of lipid and its derivatives. **CO3**: Identify the structure and general pathways of amino acids metabolism **CO4**: Illustrate the integration of metabolic pathways of energy metabolism
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 **Course Content**

UNIT I

• Biochemistry, its scope and importance in foods and nutrition.
• Carbohydrates: Digestion, absorption and metabolism of carbohydrates. Anaerobic and aerobic pathways, glycolysis, citric acid cycle, gluconeogenesis, pentose-phosphate pathway, glycogenesis, glycogenolysis, glucose transport, metabolism of lactate and pyruvate (in brief).

UNIT II

• Fats and Lipids: Digestion, absorption and metabolism of fats and fatty acids, beta-oxidation of fatty acids, biosynthesis of fatty acids, energy yield from dietary fats, storage, mobilization of fat stores during exercise, production of ketone bodies and ketogenic diets, metabolism of phospholipids, glycolipids and cholesterol (in brief).
• Proteins: Digestion, absorption and metabolism, urea cycle (in brief).

UNIT III

• Energy: BMR, Factors affecting BMR, energy requirements for physical activity, relative body weight and influence of physical exercise on changes in body fat and body composition, utilization of energy by muscle tissue in relation to exercise type, intensity and duration.
• Hormones: Classification, general mode of action, hormones of pituitary, thyroid, parathyroid, adrenals, reproductive glands and pancreas. Hormonal disorders, counter regulatory hormones.

UNIT IV

• Enzymes: Definition, classification, co-enzymes, their functions and factors affecting enzyme activity.
• Acid base balance: Concepts and disorders, pH, buffers, acidosis, alkalosis.
• Water and electrolyte balance. Role of water.

Practicals Work

1. Preparation of standards solutions Measurement of pH.
2. Carbohydrates: Quantitative estimation of sugars in foodstuff.
3. Proteins: (i) Quantitative estimation of amino acids by Ninhydrin Method. (ii) Estimation of proteins by Lowry method and Biuret method (iii) Estimation of nitrogen by Khedjals Method
4. Lipids: (i) Quantitative estimation of cholesterol (ii) Isolation and estimation of total lipids (iii) Quantitative estimation of Phospholipids.

13 **Suggested Resources for Reading**

1. Rao, K.R., Textbook of Biochemistry, Prentice Hall of India Pvt. Ltd. ND- 110001.
2. Murray, R.K. and Rodwell, V.W., Harper‘s Biochemistry, Prentice Hall of Australia Pvt. Ltd.
3. Lehninger, A.L., Principles of Bio-Chemistry, CBS Publishers and Distribution.
4. Qrten, J.M. and Neuhans, O.W., Human Bio-Chemistry, The C.V. Mosby Co., Toronto, London.

Note: Latest edition of text books may be used.

1	Course Title	Nutraceutical and Food Additives
2	**Course Code**	**NAD 302**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Product Development Manager
8	**Course Objectives**	To learn about food components and their nutraceutical importance To learn the role of food additives
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1**. learn some important food components and their nutraceutical importance **CO2**. Understand the mechanism of action of some phytochemicals and zoochemical in the treatment and prevention of diseases such as cancer, cardiovascular diseases and so on. **CO3**. Know Indian foods and some of their claimed nutraceutical properties. **CO4**. Learn the role of food additives in processing food
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 Course Content

Unit I

- Nutraceuticals: Definition, Classification, food and non-food sources, mechanism of action. Role of omega-3, fatty acids, carotenoids, dietary fiber, phytoestrogens; gluconates,organo-Sulphur compounds as nutraceuticals. Use of nutraceuticals in traditional health sciences. Their role in preventing /controlling diseases.

Unit II

- Prebiotics and Probiotics: Usefulness of probiotics and prebiotics in gastrointestinal health and other benefits. Beneficiary microbes; prebiotic ingredients in foods; types of prebiotics and their effects on gut microbes.

Unit III

- Definitions, nutritional and non-nutritional food additives, uses and functions of Acid, Base, Buffer systems, Salts and chelating / sequestering agents.

Unit IV

- Food colour, pigments, their importance and utilization, Flavoring agents and related substances
- Antioxidants, Emulsifying and stabilizing agents, Anti-caking agents, Humectants, thickeners, Firming agents., Clarifying agents.

Practical Work

1. Identification of various nutraceuticals and functional foods available in the market
2. Preparation and sensory evaluation of probiotic/prebiotic/symbiotic foods
3. Preparation and sensory evaluation of antioxidant dietary fiber rich foods.
4. To conduct the market survey for identification of health claims of various nutraceuticals products.
5. Preparations of some traditional, fermented, functional and other products.

13 Suggested Resources for Reading

1. Colleen C., Kerry G., Keith R., Salter Venzon D., Samantha I. (2012). Phytochemicals: Health Promotion and Therapeutic Potential. 1st Ed. CRC Press
2. Bagchi D., Francis C. Lau, Ghosh D.K. (2010). Biotechnology in Functional Foods and Nutraceuticals. 4th Ed. CRC press
3. Food Chemistry, O.R.Fennema
4. Food Chemistry, Belitz, Grosch
5. Food Facts & Principles by Shakuntala Manay N & Shadoksharaswamy N, 1996, New Age, World Publishers.

Note: Latest edition of text books may be used.

1	**Course Title**	**Food science**
2	**Course Code**	**NAD 303**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) **a**.The exam of this course will be of 3hours duration. **b**.There will be no objective type questions. **c**.Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. **d**.Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Scientist
8	**Course Objectives**	To provide knowledge to students for preparation of food and the scientific principle behind it**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1**: Summarize on food, food groups and various cooking methods **CO2**: understand the nutritive value and properties of cereals and pulses **CO3**: Have knowledge on composition and nutritive value of milk and various milk products **CO4**: Describe composition, nutritive value, changes during storage and role in cookery of egg, fish and flesh foods **CO5:** Distinguish between fruits and vegetables, its composition, nutritive value and effect of cooking **CO6**: Know the types, properties, nutritive value, properties and changes in cooking of sugars, fats, oils and beverages including processing of coffee, cocoa
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2..Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 Course Content

UNIT I

- Cereal and cereal products- cereals and millets, breakfast cereals and cereal products. Microscopic structure of various starch granules, nutritive value of rice, wheat and locally available millets. Processing of cereals, milling, parboiling, effect of processing and cooking on the nutritive value of cereals, dextrinization, gelatinization, gluten formation. Storage of cereals, contamination, preservation and cost.

• Pulses, legumes and nuts: Production (in brief), processing, use in a variety of preparations, some common nuts, meat substitutes- soya products, textured vegetable protein (TVP) and cost.

UNIT II

• Milk and milk products- composition, classification, properties, quality, processing, nutritive value of milk, butter, curd, butter milk, khoa, cheese and various kinds of milk–processed milk, dried milk, evaporated milk, condensed milk products.

• Vegetables and Fruits – classification, composition and nutritive value. Methods to minimize the loss of nutrients, colour, texture, flavor, browning reaction, selection, purchase, storage and availability, cost and use, nutritional aspects of raw and processed vegetables and fruits. Pectins, substance theory of pectin -gel formation, testing, factors affecting pectin and gel formation.

UNIT III

• Fats and oils-types, saturated, MUFA, PUFA, sources of fats and oil (animal and vegetables), hydrogenation, invisible fats, use of fat in cookery, factors affecting absorption of fats, smoking point, rancidity, uses, storage cost and nutritional aspects.

• Raising agents- types, constituents, use in cookery and bakery.

UNIT IV

• Convenience foods- role, types, advantages, uses, cost and contribution to diet.

• Tea, coffee, chocolate and cocoa powder, processing, use and nutritional aspects.

• Effect of pH on foods.

13 Suggested Resources for Reading

1. Charley, H.,Food Science , John Wiley& Sons
2. Griswold C., R.H., The Experimental Study of Foods,. Houghton Miffin Co.
3. Hughes, O., Introductory Foods, Macmillan & Co
4. Pan and Palamer, Food Theory and Application, John Wiley and Sons
5. Swaminathan, M., Food Science and Experimental Foods, Ganesh &Co.
6. Brain A. Fox and Allan G. Cameron, Food Science , Nutrition and Health, Edward Arnold
7. Mudambi S.R. and S.M. Rao, Food Science, Wiley Eastern LTD, New Delhi, Bangalore, Bombay, Calcutta, Madras, Hyderabad. 8. Devis, B ., Food Commodities, Heinman Professional Publishing, Great Britain.

Note: Latest edition of text books may be used.

SEMESTER – III

PW II
Project Work

- "Conduct a cost-benefit analysis of menu offerings in an institutional catering setting (e.g., a hospital or corporate canteen), optimizing meal choices for both nutritional value and budget efficiency, while maintaining food quality and patient satisfaction.

- Submission of project report.

- Viva

Semester IV

1	**Course Title**	**Human Nutrition**
2	**Course Code**	**NAD 401**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a.The exam of this course will be of 3hours duration. b.There will be no objective type questions. c.Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d.Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritionist, Dietician
8	**Course Objectives**	To provide knowledge regarding digestion and utilization of various nutrients. Apply knowledge of mechanisms of nutrients and their regulation to explain the common deficiency disease related to nutrients
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1** Practically gain knowledge to plan diet for each stage of life according to the guidelines for dietary needs **CO2** Gain knowledge on changes during various stages of growth and development throughout lifecycle **CO3** Thorough understanding of the basis of human nutritional requirements
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory ,Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 Course Content

Unit I

- Introduction to nutrition, Planning a healthy diet, Food Components, Carbohydrates: Structure, Digestion and Absorption. Disorders related to carbohydrate. Regulation of body temperature and energy metabolism. Glycemic Index and glycemic load. Protein: Structure, Digestion and Absorption. Disorders related to protein. Regulation in food intake, nitrogen balance, protein quality amino acid requirement and imbalances, relation between energy and protein requirement.

Unit II

- Lipids: Structure, Digestion and Absorption. Disorders related to lipids- hyperlipidemia, arthrosclerosis, CHD. Lipids in health management

Unit III

- Vitamins: Classification, functions, RDA, Absorption, deficiency and toxicity of vitamins(Water and fat soluble vitamins)
- Minerals: Classification, functions, RDA, Absorption, deficiency and toxicity of minerals.(Iron, Iodine, Calcium, Phosphorus, Magnesium)

Unit IV

- Lifespan Nutrition- Nutrition assessment and intervention, pregnancy and lactation, lifespan changes in nutritional requirements. Nutrition related diseases- Nutrition and GIT disorders, Nutrition and cardio vascular diseases, Nutrition and diabetes, Nutrition and Evolution

.

13 Suggested Resources for Reading

1. Ross, A. C., Caballero, B., Cousins, R. J., Tucker, K. L., & Ziegler, T. R. (Eds.). (2014). Modern
2. Nutrition in Health and Disease, Wolters Kluwer.
3. Muller, H.G., Fundamentals of Foods and Nutrition. New Delhi, Wiley Eastern Pvt Ltd.
4. Barasi Mary, E., Human Nutrition-A health perspective. Arnold International Student‘s edition
5. Mehtab S Bamji, Text Book of Human Nutrition, Oxford and IBH Publishing Co. Pvt. Ltd.
6. Joshi S.A., Nutrition and Dietetics with Indian case studies, Mc Graw Hill Education (India) Pvt.L

Note: Latest edition of text books may be used.

1	**Course Title**	**Institutional Food Service and Personnel Management**
2	**Course Code**	**NAD 402**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Service Manager
8	**Course Objectives**	To understand the different kinds of food service units and systems**.** To understand the principles of organization and management. To gain knowledge in various areas of food production**.** To learn the principles of personnel management
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Understanding food microbiology, sanitation, and hygiene, and how to apply them in food production and service **CO2**. Learning how to design menus and food service techniques, and how to manage food service operations **CO3**. Learning about personnel management, employee benefits, and training **CO4**. Learning how to apply food services technology and operate industry equipment **CO5**. Learning how to communicate appropriately with clients, staff, and management .
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practicals Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 Course Content

UNIT I

- Types of food service institutions. Styles of food services.
- Planning, writing, types and preparation of menu and menu cards. Menu display.
- Floor planning and layout.

UNIT II

- Principles, functions and tools of management (tangible and intangible).
- Food material management: Meaning definition and importance. Selection, purchase and storage of food. Control in relation to these operations (material planning, budgeting, material standardization, inventory control).
- Store keeping: Definition, objectives, functions, factors underlying successful storekeeping, duties and responsibilities of a storekeeper, purchasing, organization, principles, procedure, systems and quality control

UNIT III

- Personnel management: Recruitment, selection and training of personnel, desirable qualities and responsibilities, work standards, productivity, supervision, performance appraisal, motivation, incentives for effective performance.
- Labour policies and legislation (personnel policies related to salaries, other emoluments, allowances, leave, uniform and other prize benefits.
- Laws affecting food service operations, union and contract negotiations.

UNIT IV

- Financial management: Cost concept, cost control, pricing, budget and accounting, book keeping.
- Types, Selection, purchase, operation, care and maintenance of equipment.

Practical Work

- Standardization of recipes of snacks and a meal for portion and cost.
- Preparation of various types of menu cards.
- Table setting, tray service of tea and snacks of tea and snacks for 20 persons.
- Serving high tea to 20 persons and a meal to 10 persons.
- Preparation and service of packed lunch.
- Planning and preparation of diets for special occasions-birthdays and festivals.
- Maintenance of accounts and record keeping.
- Visit to any two food service institutions (Hospitals, flight kitchen, hotel, restaurant, canteen, industry).
- Prepare an inventory for evaluating staff‘s personal hygiene.

13 Suggested Resources for Reading

1. Bora, P.M., Food Administration in India. A Study of an Indian State, Ajanta Publishers
2. Boella, M.J., Personnel management in the Hotel and Catering Industry, BassieandJenhis, London.
3. Kiesan, J., Controlling, Analyzing Costs in Food Service Operations, MacMillan Publishing Company.
4. Kumar, H.L., Personnel Management in the Hotel and Catering Industry. Metropolitan Book Co., Delhi.
5. Minor, L.J. and Cichy, R.F., Food Service Systems Management. Avi Publishing Co., Connectient.
6. Sethi, M. and Malhan, S., Catering Management-An Integrated Approach. Wiley Eastern Ltd.

Note: Latest edition of text books may be used

1	**Course Title**	**Meal Management**
2	**Course Code**	**NAD 403**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Meal Manager
8	**Course Objectives**	To provide knowledge about basic Principle of meal planning. Exhibit a solid foundation of techniques for food preparation, presentation and service, including competence in diet planning.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1**. Develop critical thinking skill. **CO2**.Learn skills for preparing food **CO3**. Learn to produce nutritious meals **CO4**.Know about sanitation awareness **CO5**.Learn how to plan menus **CO6**.Learn how to purchase food
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory, Interactive sessions, PPT ,Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4.All questions carry equal marks.**

12 Course Content

UNIT I

• Introduction to meal management: Balanced diet, food groups, food guide and food pyramid, exchange list.
• Basic principles of meal planning, objectives, steps in meal planning, food cost calculation.
• Indian meal patterns: Vegetarian & non-vegetarian.
• Food fads & the faulty food habits.
• Recommended dietary allowances (RDA) for Indians and factors affecting these viz., age, sex, physical activity and physiological state.

UNIT II

• Nutritional and food requirements for adult man- sedentary, moderate and heavy worker and considerations while planning diet for him.
• Nutritional and food requirements for adult woman- sedentary, moderate and heavy worker and considerations while planning diet for her.
• Nutritional and food requirements for pregnant woman and considerations while planning for her diet, with special reference to complications in pregnancy.

UNIT III

• Nutritional and food requirements for lactating mother and considerations with special reference to different periods of lactation.
• Nutritional and food requirements for an infant and considerations while planning diet:
a) Breast feeding-Advantages and dis advantages, feeding patterns and technique.
b) Infant formula
c) Weaning-Definition, time, importance and steps.
d) Introduction of supplementary food.
• Nutritional and food requirements during early childhood (toddlers and preschoolers) and considerations for planning their diet with special reference to their growth need.
• Nutritional and food requirements for school going and planning diet with special reference to importance of school lunch and snacks.

UNIT IV

Nutritional and food requirements for adolescent girl and boy and considerations while planning diet for them with special reference to their growth need and eating habits.
• Nutritional and food requirements for old age and considerations for planning their diet. **.**

Practical Work

• Plan a day‘s diet for the following age groups and calculate energy, carbohydrates, protein, fat and one important vitamin or mineral as per requirement for the given age group.
a) Adult man- sedentary, moderate and heavy worker
b) Adult woman- sedentary, moderate and heavy worker
c) Pregnant woman
d) Lactating woman
e) Infant-weaning food
f) Preschool child
g) School going
h) Adolescent girl
i) Old age
• To prepare any three weaning foods.

13 Suggested Resources for Reading

1. Guthrie, H.A. Introductory Nutrition. 6th ed. Times Mirror/Mosby College Pub. St. Louis.
2. Anderson, L., Nutrition in Health and Disease, J.B. Lippincott co. Philadelphia.
3. Recommended Dietary Intakes for Indians, I.C.M.R.
4. Guthrie, H.A., Introductory Nutrition, Times Mirror/Mosby College Publ.-St. Louis.
5. Worthington, R. & Bonnie S. et al., Nutrition in Pregnancy & Lactation, Times Mirror/Mosby College, St. Louis.

Note: Latest edition of text books may be used

1	**Course Title**	**Food Quality Management**
2	**Course Code**	**NAD 404**
3	**Typology ofCourse**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Quality Manager
8	**Course Objectives**	To understand the different principles of food quality control. To assess the food quality assurance of bakery products.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.**Learn how to apply safe food handling techniques, water and waste management, and the importance of food laws and regulations **CO2.**Understand the role of food packaging and the importance of nutrition labeling **CO3.**Learn the basic principles of food preservation methods **CO4.**Learn about microbial contamination of food, spoilage, and preservation methods against microbes **CO5.**Learn about foodborne infections and intoxication, sanitation, and hygiene in food industries **CO6**.Learn about different techniques and standard procedures that are followed in industry **CO7.**Learning different certifications required for the food industry
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.The syllabus of this paper has been divided into FOUR units.** **2.Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3.The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**

4.All questions carry equal marks.

12 Course Content

UNIT I

- Objectives, importance and functions of quality control.
- Principles of food quality control and quality assurance, quality control and assessment in food materials used in baking industry and finished bakery products.

UNIT II

- Total quality management (TQM) - good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control.
- Microbial quality control: determination of microorganisms in foods by cultural, microscopic, physical, chemical, immunological and bioassay methods

UNIT III

- Food regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO9000 series etc.
- Food laws and standards, Food standards and safety Act: salient provisions and prospects, role of various national and international agencies**.**

UNIT IV

- Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of bakery products

13 Suggested Resources for Reading

1. Alli, I., Food Quality Assurance Principles and Practices, CRC Press.
2. Vasconcellos, J.A., Quality Assurance for the Food Industry. A Practical Approach, CRC Press

Note: Latest edition of text books may be used.